

RAAHIM SALMAN

BIostatistician
Health Data Analyst
Dentist

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EDUCATION

M.Sc. Statistics and Data Science

UHasselt, Belgium | Biostatistics

- Statistical Modelling
- Survival Analysis
- Epidemiology
- R & SAS
- Machine Learning

Bachelor of Dental Surgery (BDS)

Clinical background in dentistry

SKILLS

Statistical Programming

R/RStudio, SAS, SPSS, SQL, Excel, Microsoft Office

Analysis & Modelling

Linear & logistic regression, Cox PH, survival analysis, GLM/GLMM, multivariate methods, Bayesian methods

Research & Reporting

Study design, literature review, qualitative research, scientific writing, publication support

Tools

NVivo, EndNote, Overleaf

LANGUAGES

English — Fluent

Dutch — A2 (PCVO ongoing classes)

Urdu — Native

Biostatistician with a background in dentistry and clinical research, combining healthcare expertise with **statistics, data science, and machine learning**. Experienced in statistical modelling, health data analysis, study design, and scientific research using R, SAS, and SPSS. I have contributed to internationally funded research projects involving diverse populations and multidisciplinary teams, with experience in data collection, data validation, analysis, research coordination, and scientific reporting. I have also authored several peer-reviewed publications and am particularly interested in applying statistical and machine-learning methods to real-world healthcare data to support evidence-based research and decision-making.

PROFESSIONAL EXPERIENCE

Senior Research Assistant — Clinical & Public Health Research AKU

Integrated Disaster preparedness and Response program | Aga Khan University, Karachi, Pakistan | February 2024 – June 2024

- Supported clinical and public-health research through literature review, data collection, research coordination, scientific writing, and study documentation, including activities within the Department of Pediatrics and Child Health.
- Conducted qualitative and quantitative analyses, including independent coding, thematic analysis, data validation, and pre-/post-training evaluation; presented findings to principal investigators and research teams.
- Contributed to grant proposals, concept notes, pilot-study activities, and manuscript development, including qualitative research on perceptions of child abuse among teachers in Pakistan.
- Coordinated and delivered a five-day field training program for healthcare providers on disaster preparedness and response, supporting participant training, evaluation, and research-related activities.

Research Coordinator *Sehatmand Khandan Project (SMK) | Aga Khan University, Karachi, Pakistan | August 2023 – December 2023*

- Led data collection across multiple regions, including 160 in-depth interviews and 80 focus group discussions, while training data collectors and monitoring data quality.
- Coordinated participant recruitment, field validation, stakeholder engagement, and ethical/IRB compliance.
- Managed data entry, cleaning, storage, documentation, transcription, translation, and qualitative coding.
- Supported quantitative analysis and ensured analyses aligned with study objectives and hypotheses.
- Managed field logistics, research budgets, travel, staffing, contracts, and payments.
- Delivered training for data collectors and contributed to study planning, research questions, objectives, hypotheses, and project scheduling

Research Coordinator *(VOLUNTEER JOB) | Aga Khan University, Karachi, Pakistan | April 2022 – July 2023*

- Conducted literature reviews, manuscript writing, and article submissions across public health and clinical research projects.
- Supported mixed-method studies involving quantitative and qualitative data collection, analysis, and reporting.
- Contributed to projects on male infertility, reproductive health, COVID-19, gender-based violence, and elderly fall prevention.
- Prepared research reports, presentations, project documentation, study summaries, and closure reports.

INTERESTS

Biostatistics • Clinical Research •
Health Data Science • Epidemiology •
Evidence-Based Healthcare

- Participated in training on multisectoral prevention and response to gender-based violence in conflict-affected populations.

Dental House Officer *Altamash Institute of Dental Medicine, Karachi* | September 2022 – August 2023

- Participated in training on multisectoral prevention and response to gender-based violence in conflict-affected populations.
- Conducted clinical research, including proposal development, ethical approval support, data collection, analysis, manuscript writing, and article submission.
- Presented research on digital vs conventional impression techniques and contributed to studies on ChatGPT in dental education and dental anxiety.
- Completed rotations in oral surgery, prosthodontics, endodontics, and periodontics, performing extractions, restorative procedures, root canal treatment, scaling, polishing, and root planing.
- Gained hands-on experience in patient assessment, treatment planning, postoperative care, and prosthetic rehabilitation.

PUBLICATIONS

Awareness, Usage, and Perspectives on ChatGPT in Dental Education *Scientific Reports (Nature Portfolio)*, 2025

- First author; study design, data collection, SPSS analysis, regression modelling, and manuscript development

Child Domestic Labour in Pakistan: A Narrative Review *Journal of the Pakistan Medical Association*, 2025

- First author; concept, design, drafting, revision, and evidence synthesis.

The Elderly's Limited Awareness of Advanced Dental Procedures and Unmet Dental Needs in Pakistan *Journal of the College of Physicians and Surgeons Pakistan*,

- First author; research communication and manuscript development.

Impact of Dental Handpiece Noise on Anxiety and Heart Rate *BMC Oral Health*, 2024

- First author; clinical dental research, study design, data collection, and original manuscript writing.

The Collaborative Role of Psychologists with Dentists in Enhancing Dental Treatment for Female Patients *Pakistan Oral & Dental Journal*, 2024

- First author; interdisciplinary dental and psychosocial research communication.

Ethnicity, Age, pH, and Sperm Concentration in Male Infertility *Journal of Community Medicine and Public Health Reports*, 2024

- Contributed to literature review, manuscript preparation, and article submission.

Comparison of Two Surgical Approaches for Mandibular Angle Fractures *Altamash Journal of Dentistry and Medicine*, 2023

- First author; randomized clinical research in oral and maxillofacial surgery.

FUNDED/NON-FUNDED PROJECTS

Master's Thesis — Divide-and-Combine Survival Analysis for Clustered Data

- Applied Cox proportional hazards models, robust variance estimation, copula-based dependence modelling, and comparative analysis in R.
- Evaluated statistical accuracy, dependence estimation, and computational efficiency across multiple data-splitting strategies.

Clinical & Public Health Research Projects — Aga Khan University

- **Integrated Disaster Preparedness and Response Program:** supported healthcare-worker training, data collection, pre/post evaluation, analysis, and reporting.
- **Sehatmand Khandan Project:** contributed to reproductive and sexual-health research through field coordination, qualitative data collection, data management, and analysis.
- **Fall Prevention and Management Program:** contributed to mixed-method research involving data collection, analysis, reporting, and project documentation

PROFESSIONAL COURSES

- SUMMARY STATISTICS IN PUBLIC HEALTH | *Johns Hopkins University* | *Coursera* | *December 2023*
- DESIGN AND CONDUCT OF CLINICAL TRIALS | *Johns Hopkins University* | *Coursera* | *December 2023*
- CLINICAL TRIALS ANALYSIS, MONITORING, AND PRESENTATION | *Johns Hopkins University* | *Coursera* | *November 2023*
- CREATE A PROJECT MANAGEMENT TRACKER USING MICROSOFT EXCEL | *Coursera* | *November 2023*
- QUALITATIVE RESEARCH DESIGN | *Emory University* | *Coursera* | *November 2023*
- INTRODUCTION TO STATISTICS AND DATA ANALYSIS IN PUBLIC HEALTH | *Imperial college London* | *Coursera* | *November 2023*
- LINEAR ALGEBRA: LINEAR SYSTEM AND MATRIX EQUATION | *John Hopkin University* | *Coursera* | *November 2023*

CERTIFICATIONS

RESEARCH BASED

- RESEARCH METHOD IN DENTISTRY | Department of Surgery, Dentistry and Department of Continuing Professional Development | *Aga Khan University* | *ACCME- CME Activity code: DTL-1011301-21st July 2023* | *AMA PRA Category-1 Credit Hours – 3.5, July-2023*
- MANUSCRIPT WORKSHOP by Dr. Sohail Lakhani | Research Development and Review Cell (RDRC) | *Altamash Institute of Dental Medicine, Mar-2023*
- SECOND ANNUAL RESEARCH DAY | Certificate of Organizer | *Altamash Institute of Dental Medicine, Dec-2022*
- FIRST ANNUAL RESEARCH DAY | *Altamash Institute of Dental Medicine, Karachi Feb-2021*

CLINICAL BASED

- CLINICAL COMPARISON OF DIGITAL AND CONVENTIONAL IMPRESSION TECHNIQUES REGARDING FINISH LINE AND IMPRESSION TIME – Paper presentation | *Altamash Institute of Dental Medicine* | *Feb-2023*
- WORLD ORAL HEALTH DAY SYMPOSIUM (WOHDS) | Department of Surgery | *Department of Continuing Professional Education (DCPE)* | *Aga Khan University* | *ACCME – CME Activity Code : DNTL – 1011257* | *AMA PRA Category-1 Credit Hours – 03, Mar-2023*
- BIOFILM | *Angio* | *ilearn* | *Enhanced Continuing Professional Development – 20 Minutes, Jan-2023*
- BLOOD-BORNE VIRUSES | *Angio* | *ilearn* | *Enhanced Continuing Professional Development – 30 Minutes, Jan-2023*
- ROLE AND RESPONSIBILITIES IN DECONTAMINATION | *Angio* | *ilearn* | *Enhanced Continuing Professional Development – 45 Minutes, Jan-2023*
- 9TH INTERNATIONAL ORTHODONTIC CONFERENCE | *Pakistan Association of Orthodontist* | *Continuing Medical Education – 16 Credit Hours* | *International College of Dentist Dec-2022*

- PAKISTAN DENTAL ASSOCIATION AWARENESS SEMINAR | *Pakistan Dental Association | Altamash Institute of Dental Medicine | Continuing Professional Development – 06 Credit Hours | ICCDE, Dec-2022*
- A ROAD TO QUALITY POSTGRADUATE TRAINING | Postgraduate Medical Education | *Aga Khan University | ACCME- Credential ID PGME-147-2022 | Continuing Medical Education – 3 Credit Hours, Dec-2022*
- DEVELOPING PROFESSIONAL SKILLS | Angio | ilearn | *Continuing Professional Development – 30 Minutes Credit, Nov-2022*
- ENHANCED CPD SCHEME | Angio | ilearn | *Continuing Professional Development – 45 Minutes Credit –, Nov-2022*
- EVIDENCE-BASED DECISION MAKING: INTRODUCTION AND FORMULATING GOOD CLINICAL QUESTION | *Dental-care | ADA CERP, PACE Program Provider, AADH Program Provider, Continuing Education Credit – 2 Credit Hours, Nov-2022*
- EVIDENCE-BASED DENTISTRY- WHY DO I NEED THAT? | *Dental Care | ADA CERP, PACE Program Provider, AADH Program Provider, Continuing Education Credit – 1 Credit Hour | Nov-2022*
- ORAL CANCER, EARLY DETECTION OF MOUTH CANCER | Angio | ilearn | *Enhanced Continuing Professional Development – 1 Hour, Nov-2022*
- ORAL CANCER, ANATOMY AND PATHOLOGY | Angio | ilearn | *Enhanced Continuing Professional Development – 1 Hour, Nov-2022*
- ORAL CANCER, SIGN AND SYMPTOM OF ORAL CANCER | Angio | ilearn | *Enhanced Continuing Professional Development – 1 Hour –, Nov-2022*
- DECONTAMINATION OF DENTAL INSTRUMENT | Angio | ilearn | *Enhanced Continuing Professional Development – 1 Hour –, Nov-2022*
- IMMEDIATE FUNCTIONAL LOADING IMPLANTOLOGY | *International Academy of Cortical Implantology, Nov-2022*
- CLEAR PATH ORTHODONTICS | *Royal Academy of Facial Aesthetics, Karachi, July-2022*
- AKUPI-NCDS Research Training Program | *Aga Khan University | May-2022*
- YOUNG LEADER AND ENTREPRENEURS SUMMIT PARTICIPATION | *Lahore University of Management Science, Sep-2016*

REFERENCES

Dr. Naseer Ahmed,

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